

Ex 21.2

Q1 · $\frac{4x-3}{(x+1)^2}$

Let $\frac{4x-3}{(x+1)^2} = \frac{A}{(x+1)} + \frac{B}{(x+1)^2}$ — (A)

M. by $(x+1)^2$ on both sides

$$\frac{\cancel{(x+1)^2}(4x-3)}{\cancel{(x+1)^2}} = \frac{A\cancel{(x+1)^2}}{\cancel{(x+1)}} + \frac{B\cancel{(x+1)^2}}{\cancel{(x+1)^2}}$$

$$4x-3 = \overbrace{A(x+1)} + B \quad \text{--- (1)}$$

$$\text{put } x = -1 \quad \text{--- (1)}$$

$$4(-1)-3 = A(\cancel{-1+1}) + B$$

$$-4-3 = A(0) + B$$

$$-7 = B$$

$$\boxed{B = -7}$$

$$x+k=0$$
$$\boxed{x=-1}$$

Simplify eq ①

$$\underline{4x} - 3 = \underline{Ax} + A + B$$

Compare Coeff of x

$$4 = A$$

$$(A = 4)$$

poly $A=4$, $B=-7 \sim \textcircled{A}$

$$\frac{4x-3}{(x+1)^2} = \frac{4}{(x+1)} - \frac{7}{(x+1)^2}$$

✓

Q2 $\frac{x^2+7x+3}{x^2(x+3)}$

Let

$$\frac{x^2+7x+3}{x^2(x+3)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+3} - \textcircled{A}$$

M by $x^2(x+3)$. . .

$$\frac{\cancel{x^2(x+3)}(x^2+7x+3)}{\cancel{x^2(x+3)}} = \frac{A\cancel{x^2}(x+3)}{x} + \frac{B\cancel{x^2}(x+3)}{x^2} + \frac{C\cancel{x^2}\cancel{(x+3)}}{(x+3)}$$

$$(x^2 + 7x + 3 = \widehat{A(x(x+3))} + \widehat{B(x+3)} + Cx^2) \quad \text{--- (1)}$$

put $x = -3$ in (1)

$$(-3)^2 + 7(-3) + 3 = A(-3)(\cancel{-3+3}) + B(\cancel{-3+3}) + C(-3)^2$$

$$9 - 21 + 3 = 0 + 0 + 9C$$

$$-9 = 9C$$

$$\frac{-\cancel{9}}{\cancel{9}} = C$$

$$C = -1$$

fully $x=0$ is ①

$$0+7(0)+3 = A(0)(0+3) + B(0+3) + C(0)^2$$

$$3 = 3B$$

$$\frac{3}{3} = B$$

$$1 = B$$

$$\boxed{B=1}$$

Simplify eq (1)

$$\underline{x^2 + 7x + 3} = \underline{A}x^2 + 3Ax + Bx + 3B + \underline{C}x^2$$

Compare coeff of x^2

$$1 = A + C$$

put $C = -1$

$$1 = A - 1$$

$$1 + 1 = A$$

$$2 = A$$

$$(A = 2)$$

putting $A=2$, $B=1$, $C=-1$ we get $\textcircled{A}$

$$\frac{x^2+7x+3}{x^2(x+3)} = \frac{2}{x} + \frac{1}{x^2} - \frac{1}{(x+3)}$$

[Signature]

Q3 $\frac{5x^2 - 30x + 44}{(x-2)^3}$

Let

$$\frac{5x^2 - 30x + 44}{(x-2)^3} = \frac{A}{(x-2)} + \frac{B}{(x-2)^2} + \frac{C}{(x-2)^3} \quad \leftarrow \textcircled{A}$$

M. by $(x-2)^3$ -----

$$\frac{\cancel{(x-2)^3}(5x^2 - 30x + 44)}{\cancel{(x-2)^3}} = \frac{A(x-2)^{\cancel{3}^2}}{\cancel{(x-2)}} + \frac{B(x-2)^{\cancel{3}^1}}{\cancel{(x-2)^2}} + \frac{C(x-2)^{\cancel{3}}}{\cancel{(x-2)^3}}$$

$$Sx^2 - 30x + 44 = A(x-2)^2 + \widehat{B(x-2)} + C \quad \text{--- (1)}$$

$$\text{Put } x=2 \text{ in (1)}$$

$$S(2)^2 - 30(2) + 44 = A(\cancel{2-2})^2 + B(\cancel{2-2}) + C$$

$$20 - 60 + 44 = 0 + 0 + C$$

$$4 = C$$

$$\boxed{C = 4}$$

Simplify eq (1)

$$5x^2 - 30x + 44 = A(x^2 - 4x + 4) + Bx - 2B + C$$

$$5x^2 - \underline{30x} + 44 = Ax^2 - \underline{4Ax} + 4A + \underline{Bx} - 2B + C$$

Compare Coeff of x^2

$$5 = A$$

$$\boxed{A = 5}$$

Compare Coeff of x

$$-30 = -4A + B$$

put $A = 5$

$$-30 = -4(5) + B$$

$$-30 = -20 + B$$

$$-30 + 20 = B$$

$$-10 = B$$

$$B = -10$$

Put $A = 5, B = -10, C = 4$
w/ (A)

$$\frac{5x^2 - 30x + 44}{(x-2)^3} = \frac{5}{(x-2)} - \frac{10}{(x-2)^2} + \frac{4}{(x-2)^3}$$

A